

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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adoption in the literature. We interpret this assumption as if units do not “forget” about the treatment experience.⁴

Define G as the time period when a unit first becomes treated. Under Assumption 1, for all units that eventually participate in the treatment, G defines which “group” they belong to. If a unit does not participate in any time period, we arbitrarily set $G = \infty$. We define G_g to be a binary variable that is equal to one if a unit is first treated in period g (i.e., $G_{i,g} = \mathbf{1}\{G_i = g\}$) and define C to be a binary variable that is equal to one for units that do not participate in the treatment in any time period (i.e., $C_i = \mathbf{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$). Let $\bar{g} = \max_{i=1,\dots,n} G_i$ be the maximum G in the dataset. Next, denote the generalized propensity score as $p_{g,s}(X) = P(G_g = 1 | X, G_g + (1 - D_s)(1 - G_g) = 1)$. Note that $p_{g,s}(X)$ indicates the probability of being first treated at time g , conditional on pre-treatment covariates X and on either being a member of group g (in this case, $G_g = 1$) or a member of the “not-yet-treated” group by time s (in this case, $(1 - D_s)(1 - G_g) = 1$). Many of our results use a specialized version of this generalized propensity score, and, henceforth, we define $p_g(X) = p_{g,\mathcal{T}}(X) = P(G_g = 1 | X, G_g + C = 1)$ which is the probability of being first treated in period g conditional on covariates and either being a member of group g or not participating in the treatment in any time period. Let $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\} \subseteq \{2, 3, \dots, \mathcal{T}\}$ denote the support of G excluding $\bar{g}$.⁵ Likewise, let $\mathcal{X} = \text{supp}(X) \subseteq \mathbb{R}^k$ denote the support of the pre-treatment covariates. Finally, for a generic $\delta \geq 0$, let $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, \dots, \mathcal{T}\}$.

Next, we set up the potential outcomes framework. Here, we combine the dynamic potential outcomes framework of Robins (1986, 1987) with the multi-stage treatment adoption setup discussed by Heckman et al. (2016); see also Sianesi (2004). Let $Y_{i,t}(0)$ denote unit i ’s untreated potential outcome at time t if they remain untreated through time period $\mathcal{T}$; i.e., if they were not to participate in the treatment across all available time periods. For $g = 2, \dots, \mathcal{T}$, let $Y_{i,t}(g)$ denote the potential outcome that unit i would experience at time t if they were to first become treated in time period g . Note that our potential outcomes notation accounts for potential dynamic treatment selection, though it also accommodates (pre-specified) treatment regimes (Murphy et al., 2001; Murphy, 2003). The observed and potential outcomes for each unit i are related through

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \cdot G_{i,g} \quad (2.1)$$

In other words, we only observe one potential outcome path for each unit. For those that do not participate in the treatment in any time period, observed outcomes are untreated potential outcomes in all periods. For units that do participate in the treatment, observed outcomes are the unit-specific potential outcomes corresponding to the particular time period when that unit adopts the treatment.

We also impose the following random sampling assumption.

Assumption 2 (Random Sampling). $\{Y_{i,1}, Y_{i,2}, \dots, Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, \dots, D_{i,\mathcal{T}}\}_{i=1}^n$ is independent and identically distributed (*iid*).

Assumption 2 implies that we have access to panel data; our results extend essentially immediately

⁴See Han (2020), de Chaisemartin and D’Haultfoeulle (2020) and Bojinov et al. (2020) for alternative setups where treatment can “turn off”.

⁵When there is a “never treated” set of units with $G = \infty$, $\mathcal{G}$ only excludes this group. When such “never-treated” group is not available, we exclude the latest-treated group as there will be no valid untreated comparison group for them.

Theorem 1. *Let Assumptions 1, 2, 3 and 6 hold.*

(i) *If Assumption 4 holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,*

$$ATT(g, t) = ATT_{ipw}^{nev}(g, t; \delta) = ATT_{or}^{nev}(g, t; \delta) = ATT_{dr}^{nev}(g, t; \delta).$$

(ii) *If Assumption 5 holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $g - \delta \leq t < \bar{g} - \delta$,*

$$ATT(g, t) = ATT_{ipw}^{ny}(g, t; \delta) = ATT_{or}^{ny}(g, t; \delta) = ATT_{dr}^{ny}(g, t; \delta).$$

Theorem 1 is the first main result of this paper. It provides powerful identification results that extend the DiD identification results based on the outcome regression approach of Heckman et al. (1997, 1998), the IPW approach of Abadie (2005), and the DR approach of Sant’Anna and Zhao (2020) to the multiple-periods, multiple groups setup. In other words, Theorem 1 says that, from an identification point of view, one can recover the $ATT(g, t)$ ’s by exploiting different parts of the data generating process: the OR approach only relies on modeling the conditional expectation of the outcome evolution for the comparison groups, the IPW approach relies on modeling the conditional probability of being in group g , whereas the DR approach exploits both OR and IPW components.

In order to extend the results of Heckman et al. (1997, 1998), Abadie (2005), and Sant’Anna and Zhao (2020) to the multiple groups, multiple periods framework, we have to address two different challenges: one associated with an appropriate reference time period and one associated with an appropriate comparison group. Theorem 1 highlights how a solution to these challenges is directly connected to the limited anticipation and the conditional parallel trends assumptions. More specifically, Theorem 1 says that we can use the time period $t = g - \delta - 1$ as an appropriate reference time period under Assumption 3 and either Assumption 4 or 5. This is the most recent time period when untreated potential outcomes are observed for units in group g . Interestingly, the more treatment anticipation is allowed (i.e., the higher δ is), the further back in time one needs to go.⁹ Theorem 1 also suggests that the choice of comparison group is directly tied to the conditional parallel trends assumption one makes: under Assumption 4, one can use “never treated” units as a fixed comparison group for all “eventually treated” units; whereas, under Assumption 5, one can use the “not-yet-treated by time $t + \delta$ ” units as a valid comparison group for those who are first treated at time g . In this latter case, Theorem 1 also highlights that when all units eventually gets treated ($\bar{g} < \infty$), one is only able to identify the $ATT(g, t)$ ’s for time periods before the last treated group “effectively” starts their treatment, i.e., $t < \bar{g} - \delta$. In this case, one can not identify the $ATT(g, t)$ for the last treated cohort, too.

Finally, we note that when pre-treatment covariates play no role in identification (i.e., Assumptions 3, 4, and 5 hold unconditionally on X), (2.2)-(2.4) collapse to

$$ATT_{unc}^{nev}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | C = 1], \quad (2.8)$$

and (2.5)-(2.7) collapse to

$$ATT_{unc}^{ny}(g, t; \delta) = E[Y_t - Y_{g-\delta-1} | G_g = 1] - E[Y_t - Y_{g-\delta-1} | D_{t+\delta} = 0]. \quad (2.9)$$

⁹As mentioned in Remark 1, as one allows δ to increase, Assumptions 4 and 5 becomes more restrictive.